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Source-independent quantum random number generators with integrated silicon photonics

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Random numbers play a crucial role in numerous scientific applications. Source-independent quantum random number generators (SI-QRNGs) can offer true randomness by leveraging the fundamental principles of quantum mechanics, eliminating the need for a trusted source. Silicon photonics demonstrates significant promise for QRNG due to its benefits in miniaturization, cost-effective device manufacturing, and compatibility with CMOS microelectronics. This study experimentally demonstrates a silicon-based discrete variable SI-QRNG. Our SI-QRNG system achieves a low error rate of only 0.21%, thanks to the inherent stability of the silicon-based decoder chip and its excellent polarization extinction ratio. Additionally, by using a laser with a higher repetition rate and a robust simulation model, we achieve the highest quantum random number generation rate of 9.49 Mbits per second. Our research paves the way for integrated SI-QRNGs, providing a cost-effective and robust secure QRNG module for next-generation communications.

Random numbers are crucial in daily life and scientific research. Ideally, a random number generator (RNG) should generate a sequence of numbers that are both uniformly distributed and unpredictable[1]. However, pseudo- or classical RNGs, which rely on deterministic computational algorithms or classical physical processes, exhibit predictability and long-range correlation, which cause errors in scientific simulations[2]. Such pseudo-randomness can lead to catastrophic errors in scientific simulations, particularly in fields such as secure communication[3,4], lotteries, and blockchain technology, where high information security and privacy are essential. In contrast, quantum random numbers[5–7] arise from the inherent randomness of quantum mechanics, offering random numbers characterized by both uniformity and unpredictability.

Quantum random number generators (QRNGs) have seen decades of development and experimental implementation using various quantum sources. These sources are primarily based on single photons[8,9], laser phase fluctuations[10–12], vacuum states[13–15], and path-entangled quantum states[16,17]. However, imperfections in QRNG systems will result in deviations from the ideal models used in security analysis. These deviations can lead to the risk of side information leakage, making the generated random numbers predictable.

Based on the level of trust devices, QRNGs can be classified into three main types. The first type is fully trusted QRNGs, which assumes both the source and devices are trusted and can generate high-speed random numbers[18–20]. The second type is device-independent QRNGs[21–25], which generate true random numbers without making any assumptions about the sources and measurement devices. However, this type of QRNG requires loophole-free Bell tests, resulting in low rates in practical applications. The last type is semi-device-independent QRNGs, which require reasonable assumptions to bound the side information. These assumptions relate to sources[26–28], measurement devices[29–31], the dimension of the underlying Hilbert space[27,32], and the energy consumption constraint[33–37] to achieve reasonably fast and secure random number generation.

Source-independent QRNGs (SI-QRNGs)[38,39] is a representative scheme of semi-device-independent QRNGs. This scheme requires the measurement devices to be trusted and characterizable without relying on any assumptions about the source. The core idea of SI-QRNGs is to use measurements for source monitoring, which is usually more challenging to characterize than the measurement devices themselves. In such situations, it's necessary to randomly switch between various measurement settings, often using complementary settings, to thwart the source (assumed to be under the control of an adversary)

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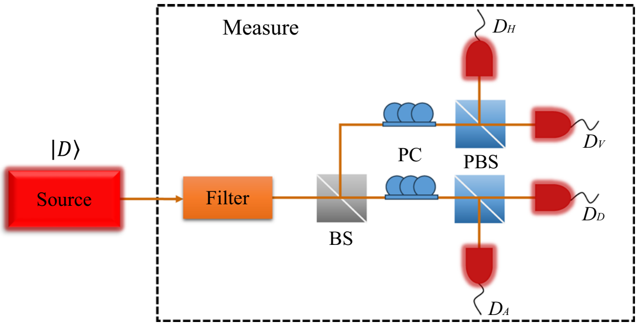

Fig. 1 | The schematic of an SI-QRNG protocol. It includes an untrusted source and a trusted measurement device. The measurement section consists of components Filter, optical filter; BS, beam splitter; PC, polarization controller; PBS, polarizing beam-splitter; D_H (D_V, D_D, D_A), the threshold detectors.

from predicting the subsequent measurement. The feasibility of SI-QRNGs has been experimentally demonstrated using bulk optical devices[39–41]. Substantial efforts have been dedicated to improving performance and practical security[42–48]. Similar to trusted-device QRNGs[49–55], integrated photonics are employed in SI-QRNG to reduce both size and cost. Remarkably, the random number generation rate reaches Gbps when using continuous-variable states[56]. However, to the best of our knowledge, an integrated discrete-variable SI-QRNG has not yet been reported.

In this study, we present a discrete-variable SI-QRNG implemented on a silicon integrated chip. The system features a straightforward structure, mainly comprising a laser diode (LD) chip, a polarization decoder chip with integrated polarization tracking, and single photon detectors (SPDs). By calibrating the chip well and optimizing parameters based on the protocol in ref. 40, our chip-based QRNG achieved a quantum bit error rate (QBER) of 0.21% and generated random numbers at a rate of 9.49 Mbits per second (Mbps). The final random numbers passes all NIST test suite criteria. The experimental results of this study illustrate the feasibility of implementing SI-QRNG with silicon integrated devices, offering a cost-effective, robust, and scalable integrated QRNG module for next-generation secure communication, the Internet of Things, and scientific simulations.

Results

SI-QRNG protocol

We first briefly review the SI-QRNG protocol[39]. A schematic of the SI-QRNGs protocol is shown in Fig. 1. In this setup, an untrusted randomness source emits light pulses. Subsequently, a trusted measurement device performs projection measurements on these pulses, leading to the generation of a random sequence. The operational details of the SI-QRNGs protocol are as follows:

1. **Source:** An untrusted random source is used as the state preparation device to prepare a polarization state $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$, where $|H\rangle$ and $|V\rangle$ represent horizontal and vertical polarization states, respectively. Because the source is untrusted and may be controlled by eavesdroppers, the quantum state prepared has arbitrary and unknown dimensions. In this scenario, randomness can still be quantified and extracted through the following steps.
2. **Squashing:** The squashing process involves mapping quantum states of arbitrary dimensions prepared by the untrusted source into qubits or vacuum states. The vacuum components are considered to account for optical losses and quantum efficiency. In practical experiments, the squashing process is achieved by introducing a series of different types of optical filters to remove unexpected optical modes.
3. **Random sampling:** During the measurement stage, Alice implements a POVM that measures the quantum state in either the X or Z basis using a passive basis selection method. In practice, The total bit number measured is $N = N_X + N_Z$, where N_X bits are measured in the X basis and N_Z bits are measured in the Z basis. It's important to note that this protocol is loss-tolerant. In an ideal scenario, the measurement device would choose the measurement basis after confirming that the received quantum state is not in the vacuum state. However, in experimental settings, the measurement basis is typically chosen before confirming whether the state is in the vacuum state, typically by observing whether the detectors register a click. The measurement device cannot predict where the losses will occur. Therefore, the impact of losses only results in a reduction in the sizes of N_X and N_Z, but the actual positions for effectively measuring the X and Z basis remain random.
4. **Parameter estimation:** When choosing to measure in the X basis, error click events include both the response outcome of $|A\rangle = (|H\rangle - |V\rangle)/\sqrt{2}$ and half of the double click events. Let the measurement probability of the X basis be $p_X = N_X/N$. Alice detects k errors among the N_X qubits measured in the X basis. If Alice measures all qubits in the X basis, the total number of errors is m. Then, the number of errors in the qubits measured in the Z basis is given by $m - k$, which is the key parameter that needs to be determined through random sampling. The quantity $m - k = N_Z e_{pZ}$ determines the extraction rate of the random numbers. The lower bound of the phase error rate e_{pZ} is defined by $e_{pZ} \le e_{bX} + \theta$, where e_{bX} is the bit error rate of the X basis and θ is the deviation caused by statistical fluctuations. According to the random sampling results proposed by Fung et al.[57], we can bound the probability of failure of $e_{pZ} \le e_{bX} + \theta$ as

$$\varepsilon_\theta = \mathrm{Prob}\left(e_{pZ} > e_{bX} + \theta\right) \le \frac{1}{\sqrt{p_X(1-p_X)e_{bX}(1-e_{bX})N}} 2^{-N\xi(\theta)}, \tag{1}$$

where $\xi(\theta) = H(e_{bX} + \theta - p_X\theta) - p_X H(e_{bX}) - (1-p_X)H(e_{bX} + \theta)$. $H(x) = -x\log_2(x) - (1-x)\log_2(1-x)$ is the binary Shannon entropy function. In data post-processing, the failure probability ε_θ is typically chosen to be 2^{-100} [40]. Due to none assumptions about the source, it may emit multiphoton states. When using threshold detectors may result in the occurrence of double click events, further increasing the error rate e_{bX} in the X basis, thereby reducing the extraction of random bits.
5. **Randomness extraction:** It has been proven that using Toeplitz matrix can effectively perform phase error correction (randomness extraction)[58]. Suppose the number of qubits measured in the Z basis is N_Z. Then, the number of bits sacrificed in phase error correction is given by $N_Z H(e_{bX} + \theta)$, and the probability of failure in phase error correction is 2^{-t_e} [59]. The parameter t_e is selected by Alice to balance the failure probability and the final output length. Given all these, the number of extracted random bits is

$$R = N_z - N_z H\left(e_{bX} + \theta\right) - t_e, \tag{2}$$

with the failure probability (in trace-distance measure) is given by $\varepsilon = \sqrt{\left(\varepsilon_\theta + 2^{-t_e}\right)\left(2 - \varepsilon_\theta - 2^{-t_e}\right)}$.

Security analysis for practical setup

In the original theoretical proposal[39], the authors addressed several practical issues, such as multiphotons, loss, and double clicks. However, it lacks an analysis of coherent attacks at the source, the fair sampling loophole, and imperfections between the X and Z basis measurements. Here, we include this analysis for the practical setup.

Firstly, the security proof of the exploited SI-QRNG protocol is based on complementary uncertainty relations as detailed in ref. 39. It incorporates well-established security proof ingredients from QKD, including phase error correction[60], random sampling[57], and the squashing model[61]. Notably, phase error correction has been demonstrated to be an effective method against coherent attacks[4,60]. Therefore, the SI-QRNG scheme can resist coherent attacks, which, as a more general form of attack, include the memory effects of the source.

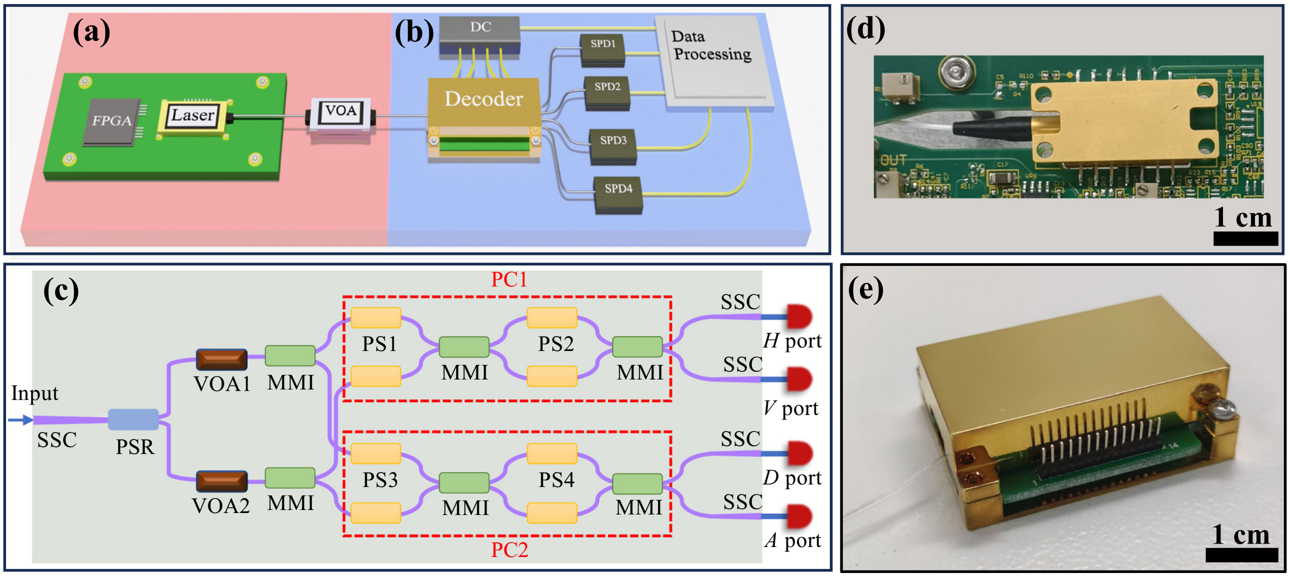

Fig. 2 | Silicon-based SI-QRNG system. a Untrusted randomness source part. Laser, a laser diode; VOA, a variable optical attenuation. **b** Measurement part. Decoder, polarization state demodulation chip; SPD, single photon detector; Data processing, including a time digital convert and a personal computer; DC, programmable DC power supply. **c** Schematic of the polarization state demodulation chip. SSC spot-size converter, PSR polarization splitter-rotator, VOA variable optical attenuator, MMI multimode interferometer, PS thermo-phase shifter; **d** the physical picture of a laser diode with using a standard butterfly package; **e** picture of the decoder chip packaged on the control board.

Secondly, a key assumption in QKD security proofs is fair sampling-the measured photon samples, even with very low detection efficiency, accurately represent the entire ensemble[62]. If a malicious eavesdropper exploits device imperfections to violate the fair sampling assumption, it could result in the disappearance of the randomness in the system's output. A well-known example of hacking a realistic QKD system by violating the fair sampling assumption is the so-called detector blinding attack. By illuminating tailored bright light into single-photon detectors, eavesdroppers can break the fair sampling assumption and then tracelessly acquire the full secret key[63]. Therefore, similar to conventional QKD, SI-QRNG must implement additional defense strategies to ensure the fair sampling assumption if we consider realistic security. Fortunately, recent works have already addressed the detector blinding attack against SI-QRNG systems[47,48].

Thirdly, the actual measurement basis X and Z are not complementary to each other, and the measurement efficiencies in the two eigenstates differ. This issue can be addressed using the security proof developed in ref. 40. In this case, the final random number generation rate under a finite regime is given by:

$$R_{\text{final}} = \frac{2\min(\eta_H, \eta_V)}{\eta_H + \eta_V}\left[-2N_Z^s \log_2 \max_{X,Z}|\langle X|Z\rangle| - N_Z^s H(e_{bX} + \theta) - t_e\right], \tag{3}$$

where η_H and η_V are the detection efficiencies of the Z-basis detectors. $2\min(\eta_H, \eta_V)/(\eta_H + \eta_V)$ is the coefficient considering the detector efficiency mismatch. Term $-2\log_2 \max_{X,Z}|\langle X|Z\rangle|$ represents the incompatibility between the Z and X measurement bases. N_Z^s is the total number of single click events on the Z-basis. In experiment, the term $-2\log_2 \max_{X,Z}|\langle X|Z\rangle|$ and the efficiency η_H and η_V can be first measured during a calibration procedure on the measure device and the detailed method of quantity estimation are provided in Methods. It is important to note that the details of the detector efficiency mismatch must remain confidential to prevent attacks similar to time-shift attacks in QKD[64].

Setup

Figure 2 shows the schematic of our chip-based SI-QRNG. This system primarily consists of two modules: a randomness source which includes an untrusted laser source, controlled by a field-programmable gate array (FPGA), as shown in Fig. 2a, and a trusted photonic polarization measurement, including a decoder and four single photon detectors (SPDs, WT-SPD300, Qasky Co. Ltd.), as illustrated in Fig. 2b. The source and measurement are interconnected through single-mode fiber links, which also function as filters. In the source, a laser with a repetition rate of 50 MHz emits pulsed light sequences at a central wavelength of 1550 nm and a pulse width of 200 ps. The laser employs a butterfly package, a standard form of optoelectronic device packaging. The packaged laser has a volume of approximately $3.00 \times 1.25 \times 0.85$ cm^3, as shown in Fig. 2d. Subsequently, the laser is attenuated by a variable optical attenuation (VOA, DA-100, OZ Optics Ltd.). In the measurement, a polarization decoder chip is employed to decode the polarization state of the photons, which are then detected by an external SPDs and recorded using a data processing module.

The central device in the SI-QRNG system is a polarization decoder chip, which can monitor the source using a conjugate basis. The schematic of the decoder chip is depicted in Fig. 2c. The chip was fabricated using a standard silicon photonics foundry and has dimensions of 1.6×1.7 mm^2. It was packaged using chip-on-board assembly, resulting in a total volume of approximately $3.95 \times 2.19 \times 0.90$ cm^3, as shown in Fig. 2e.

Any polarization state $|\psi\rangle_{pol} = \alpha|H\rangle + \beta|V\rangle$, where $\alpha^2 + \beta^2 = 1$, transmitted from the source section is initially coupled into the chip through a spot-size converter (SSC). Subsequently, it is converted into path information using a polarization splitter-rotator (PSR)[65]. Two symmetric multimode interferometers (MMIs) are used for passive selection of measurement basis in the Z or X basis.

The Z and X basis measurement sections are primarily constructed using two polarization controllers, PC1 and PC2, on the chip. Each polarization controller consists of a pair of phase shifters (PSs) connected to a Mach-Zehnder interferometer driven by another pair of PSs. Additionally, the power settings of the PSs on the decoder chip, PC1 and PC2 are adjusted to create a POVM with $|\psi\rangle_{pol}$ as the reference polarization state.

PC2 is defined as the eigenmeasurement basis for the input quantum state $|\psi\rangle_{pol}$ in the X basis, while PC1 is configured as the complementary measurement basis for the input quantum state $|\psi\rangle_{pol}$ in the Z basis. This configuration of the decoder chip leads to the measurement of the incident polarization state in the X basis and generates outputs from the D port, with detection events at the A port classified as errors.

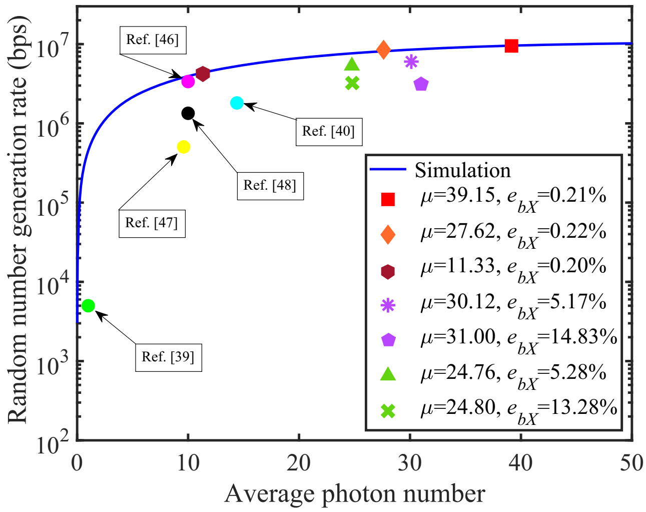

Fig. 3 | Final quantum random number generation rate versus average photon number μ. μ is defined as the average photon number before entering the detector, accounting for losses in the decoder. The blue line represents the simulation results based on our experimental system parameters. The red square represents the experimental results under an ideal case, which is (μ = 39.15, e_{bX} = 0.21%). The orange diamond represents the experimental results using a non-optimal average photon number but well-calibrated polarization, which is (μ = 27.62, e_{bX} = 0.22%). The dark red hexagon represents the experimental results obtained with a non-optimal average photon number but with well-calibrated polarization, which is (μ = 11.33, e_{bX} = 0.20%). The purple asterisk and pentagon represent the experimental results using a non-optimal average photon number and uncalibrated polarization, which are (μ = 30.12, e_{bX} = 5.17%) and (μ = 31.00, e_{bX} = 14.83%), respectively. The green triangle and cross represent the experimental results using a non-optimal average photon number and uncalibrated polarization, which are (μ = 24.76, e_{bX} = 5.28%) and (μ = 24.80, e_{bX} = 13.28%), respectively. We also plotted the highest quantum random number generation rates for different SI-QRNGs experiments (circular markers).

Furthermore, when the incident polarization state is measured in the Z basis, it generates raw random bits, which are output from either the H or V port. Lastly, the photons measured in the Z or X basis are coupled out of the decoder chip through the SSC. Detection events are collected and post-processed using a data processing module. Detailed information about the principles, fabrication process, and parameters of the polarization state decoder chip can be found in refs. 66,67.

Experimental results

The initial step involved characterizing the decoder chip, which has a total insertion loss of about 4.6 dB. Approximately 3 dB of this loss is attributed to coupling-in and -out, while the remaining 1.6 dB is due to other components and transmission losses on the chip. The phase shifters (PSs) on the chip have a 3 dB bandwidth of approximately 3 kHz and a half-wave voltage of 0.72 V. The polarization extinction ratio of the decoder chip exceeds 26 dB. Additional tests show that the probability of selecting the Z basis is 53.96%. The quantum bit error rate (QBER) for the incident polarization state in the X basis is 0.21%. The probability of a random collapse to the H port during Z basis measurement is 49.02%. To maximize the random number generation rate, we recalibrate the SPDs for higher counting rates, albeit at the expense of reduced detection efficiency. The detection efficiencies of the detectors at the H, V, D, and A ports are 1.93%, 1.93%, 1.90%, and 1.92%, respectively, with an average maximum counting rate of 8.1 MHz.

Using above characterization parameters, we can estimate the achievable random number generation rate with various incident average photon number μ for our chip-based SI-QRNG system using a theoretical simulation. The detailed theoretical model for simulation is provided in Supplementary Note 1. The results are plotted as blue line in Fig. 3. It can be seen that the highest rate is obtained when the μ approach to 39, we refer this analysis into Supplementary Note 1.

To test the performance of our setup, we control the VOA and polarization controller (PC) to obtain different average photon numbers and X-basis QBERs simulating various scenarios in SI-QRNG. The first scenario where the setup work in an optimized X-basis QBER but different optimized average photon numbers. By carefully calibrating setup, we test three cases win the following conditions: (1) μ = 11.33, e_{bX} = 0.20%; (2) μ = 27.62, e_{bX} = 0.22%; and (3) μ = 39.15, e_{bX} = 0.21%. The second scenario where the untrusted light source emitting non-optimal average photon numbers and uncalibrated polarized light. For this purpose, we adjust an additional PC at the source end of the setup shown in Fig. 2 to simulate the emission of uncalibrated polarized light. We first adjust the source sends states with different averages using the VOA and well-calibrated polarization states; additionally, we adjust the average photon number entering the detector using a VOA at the source then adjust the PC to generate non-ideal polarization states. After analyzing the detected events, we obtain various cases in the following conditions: (1) μ = 30.12, e_{bX} = 5.17%; (2) μ = 31.00, e_{bX} = 14.83%; (3) μ = 24.76, e_{bX} = 5.28%; (4) μ = 24.80, e_{bX} = 13.28%.

In each case, a total of $N' = 10^{10}$ optical pulses are sent over 200 s. By post-processing detection events, we obtain all experimental parameters necessary for estimating the random number generation rate, including the single-click counts for the detector H, denoted as N_H^s. All obtained experimental parameters are listed in Supplementary Note 2. We then calculate the secure random number generation rate under various conditions using Eq. (3) and apply the Toeplitz Matrix Hashing method[58] to derive final random numbers from the raw bits. The final secure random number rates for different conditions are presented in Fig. 3. The experimental data in Fig. 3 and Supplementary Note 2 show that in the first scenario, the obtained rate matches the simulation results. Particularly, when μ = 11.33, which approaches the maximum average photon number reported in previous works, our setup achieves a rate that exceeds all previous results. When μ = 39.15, we obtain a maximum random number generation rate of 9.49 Mbps. In the second scenario, non-optimal average photon numbers and incident polarized light lead to a reduction in raw random numbers and an increase in the bit error rate e_{bX}, further decreasing secure random numbers. Furthermore, all tests demonstrate the source-independent nature of our setup: the random number generation rate obtained under non-optimal conditions does not exceed that in optimal conditions.

For comparison, we plot the quantum random number generation rates obtained from previous SI-QRNG experiments in Fig. 3, represented by circles. Detailed data comparisons are provided in Table 1. Our results represent the highest rate, and only our work utilizes silicon-based integrated chips, as shown in the Table 1. The highest rate is attributed to the inherent stability and superior polarization extinction ratio of our decoder chip, which achieves a low error rate of 0.21%, combined with the highest repetition rate of the laser and a well-developed simulation model, yielding an almost lowest product of μ and η, a key parameter related to the final generation rate. Furthermore, we employ the same security analysis scheme as in ref. 40, enhancing the security of our chip-based QRNG compared to the original protocol[39]. Refs. 47,48 account for more comprehensive imperfection factors related to measurement devices.

We then assess the randomness of the generated random numbers by applying the NIST SP 800-22 test suite. For each case, we extract a total of 1.5 Gbits of secure random numbers and divide them into 1500 samples of 1 Mbit from the final random numbers for NIST testing. The significance level was set at α = 0.01. A test is considered passed if the P value for each NIST test term is greater than 0.01, and the corresponding proportion value falls within the confidence interval of $(1-\alpha) \pm 3\sqrt{(1-\alpha)\alpha/n} = 0.99 \pm 0.00771$. All test results are listed in Supplementary Note 3 and a visual illustration of the detailed test results for the optimal condition (μ = 39.15) are shown in Fig. 4. The P values and proportions for each test passed indicate that the experimentally obtained random numbers exhibit the expected statistical characteristics.

Table 1 | A list of discrete variable SI-QRNG experiments

Reference	Untrusted source	Uncharacterised measurement	Platform	Freedom of photon	f (MHz)	μ	e_{bX} (%)	η (%)	R (bps)	Security parameters
Cao et al.[39]	✓	×	Off-chip	Polarization	1	1	2	45	5.00×10^{3}	2×2^{-50}
Li et al.[40]	✓	×	Off-chip	Polarization	4	14.4	0.33	10	1.81×10^{6}	2×2^{-50}
Lin et al.[46]	✓	×	Off-chip	Phase	20	10	N/A	13 and 17	3.37×10^{6}	2×2^{-50}
Liu et al.[47]	✓	×	Off-chip	Polarization	5	9.6	3.5	39	5.05×10^{5}	N/A
Lin et al.[48]	✓	✓	Off-chip	Phase	20	10[a]	N/A	13 and 17	1.34×10^{6}	10^{-10}
This work	✓	×	On-chip	Polarization	50	39.15	0.21	Z basis: 1.93 and 1.93; X basis: 1.92 and 1.90	9.49×10^{6}	2×2^{-50}

f is the system repetition rate, μ is defined as the average photon number before entering the detector, accounting for losses in the decoder, e_{bX} is the quantum bit error rate of the X basis, η is the detection efficiency of the detectors, and R is the final quantum random number generation rate. [a] The estimated value from the graph of the article. Note that ref. 47,48 explore further aspects of the practical security of measurement devices.

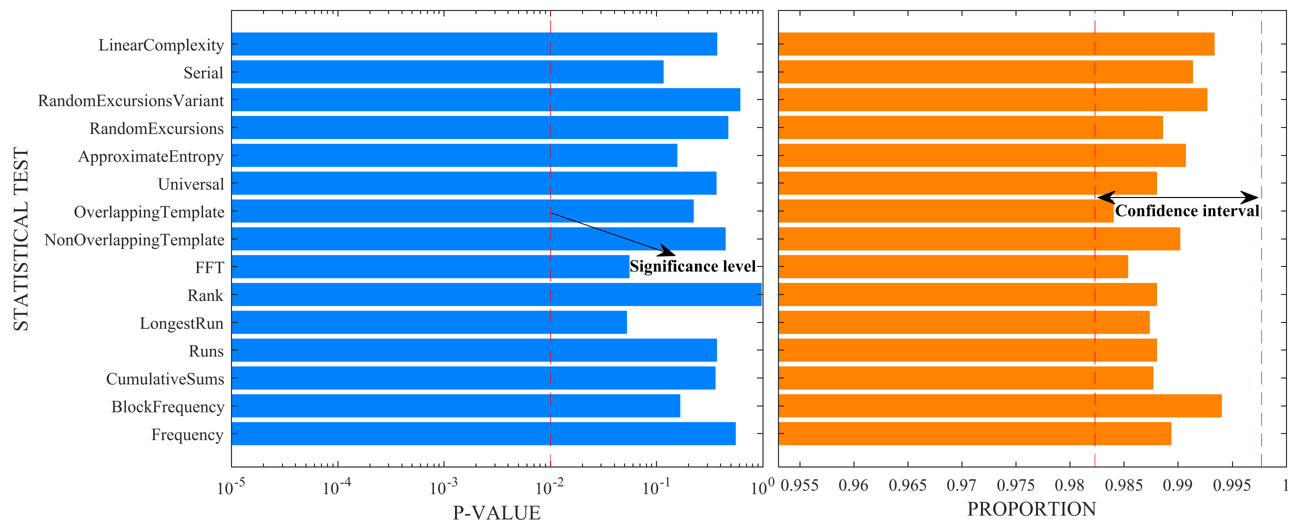

Fig. 4 | The *P* value and proportion value of NIST test terms. We subjected the random numbers collected and processed in the experiment to the standard NIST test suite using 1500 samples of 1 Mbit. In the case of multi-outcome tests, a further KS uniformity test is applied to the *P* values, and the corresponding proportions are averaged.

Discussion

In this study, we propose and experimentally demonstrate a chip-based SI-QRNG that generates random sequences by measuring photon polarization from an untrusted source. Our photon polarization decoding chip offers convenient reconfigurability, allowing for polarization state analysis and compensation without the need for additional devices. Thanks to the inherent stability and superior polarization extinction ratio of our decoder chip, the SI-QRNG system achieves a low error rate of 0.21%. This, combined with the higher repetition rate of the laser and a well-developed simulation model, enables us to achieve a final random number generation rate of up to 9.49 Mbps. We subjected the generated random numbers to NIST testing, and all tests passed. Compared to recent discrete-variable SI-QRNGs, our work strikes a balance between integration, speed, and security. Our proposed integrated SI-QRNG offers advantages such as robustness, security, and chip-level integration. It is well-suited for portable mobile platforms, airborne applications, and satellite-based quantum key distribution (QKD) applications[68].

Future research could achieve higher rates of quantum random number generation by utilizing state-of-the-art high-performance detectors[69,70] with increased counting rates. Additionally, our setup can be further integrated with germanium-on-silicon (Ge on Si) technology[71,72], a promising avenue for developing chip-scale SPD that operate at room temperature. Moreover, our setup exploits several integrated components, such as a PSR and a MMI, compared to previous studies[39,40]. Since we assume that an eavesdropper cannot exploit the realistic imperfections of the MMI in our current work, it is crucial to consider the practical security risks they may pose. For example, an eavesdropper could manipulate the operational characteristics of the MMI by adjusting the frequency or intensity of incident light.

Methods

In this section, we provide detailed experimental methods, including how to characterize the average photon number μ before detection, the detector efficiency η, and how to estimate the key term $-2\log_2 \max_{X,Z}|\langle X|Z\rangle|$. Here, μ is defined as the average photon number before entering the detector, accounting for losses in the decoder. This is because the SI-QRNG protocol's loss tolerance enables high-loss implementations by increasing the average photon number of sources to compensate for system loss.

Characterization of the average photon numbers and the detection efficiency

The method of characterizing μ and η is similar to that used in the classical single-photon detector community[73]. As shown in Fig. 5, the schematic setup is similar to that in Fig. 2, except we assume the source is trusted and use an additional VOA and a power meter to calibrate the average photon number of the incident light entering the detector. Since the typical optical power entering the single photon detector is approximately −70 dBm, which exceeds the measuring range of standard power meters (typically from 10 dBm to

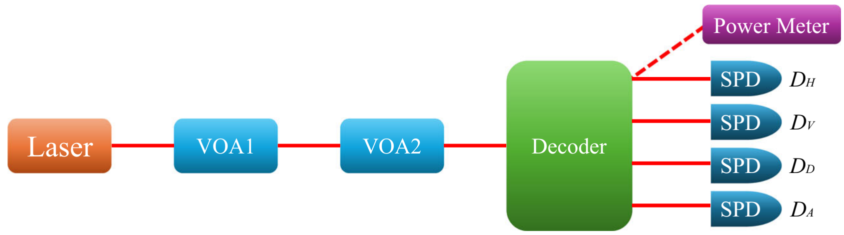

Fig. 5 | Setup of characterization average photon number and detector efficiency. VOA variable optical attenuation, SPD single photon detector.

−50 dBm), consequently, using only one VOA does not allow us to directly characterize the optical power entering the detector using the power meter. Therefore, we employ a method called the difference attenuation value method, which necessitates the use of two VOAs, to calibrate the average photon number. The detailed calibration process is as follows:

First, we use a trusted laser source, which refers to a gain-switching laser from a producer assumed to be trusted and non-hostile. Furthermore, the operational parameters of the laser (such as wavelength, intensity, and spectral purity) are well-characterized. The gain-switching laser is triggered by a 50 MHz periodic signal to generate pulsed light, which is transmitted into the decoder chip after passing through two VOAs.

Then, we scan the voltages of PS1 and PS2 until the detection counts at the H port and V port are nearly equal. Next, we scan the voltages of PS3 and PS4 until the detection count at the D port is maximized and the detection count at the A port is minimized. At this stage, the decoder chip is configured for quantum measurements in the Z and X basis.

Subsequently, we adjust the intensity of the incident light using the two VOAs. The target value of intensity of the incident light is approximately $\mu = 30$, which lies within the testing conditions to ensure that the measured detection efficiency is linear with the varying photon number. We label the corresponding attenuation values of VOA1 and VOA2 as V_1 dB and V_2 dB.

Next, we determine the light power entering the H port using the difference attenuation value method. Specifically, we replace the SPD at the H port with a power meter. We set the attenuation values of the VOA1 and VOA2 to 0, record the light power with the power meter (represented as I_0), set the VOA1 attenuation value to V_1 dB and record the light power again (represented as I_1), and then set the VOA1 attenuation value to 0 and the VOA2 attenuation value to V_2 dB, and record the light power (represented as I_2). We then calculate the light power entering the H port as $P = I_0 - I_1 - I_2$, and further deduce the average photon number $\mu_H = P/(h\nu f)$, where h is Planck's constant, ν is the frequency of the light, and f is the repetition rate of the system. We use the same method to characterize the average photon numbers for the V port, D port, and A port, recorded as μ_V, μ_D, and μ_A, respectively.

Then, we connected the four ports to the SPDs and recorded the counting rates of the four detectors, labeled as C_H, C_V, C_D, and C_A. Subsequently, we derive the detection efficiency of each detector using the formula $f(Y_0 + 1 - \exp^{-\mu_\lambda \eta_\lambda}) = C_\lambda$, where f is the system's repetition frequency, Y_0 is the detector's dark count rate, and λ represents port $\{H, V, D, A\}$. The calibrated detector efficiencies η_H, η_V, η_D, and η_A are listed in Supplementary Table 1. The average photon number before detection is calculated as $\mu = \mu_H + \mu_V + \mu_D + \mu_A$. Here, μ represents the average photon number before detection. Since losses in the decoder can always be compensated by increasing the emission intensity.

Table 2 | The raw counts of the four SPDs when the trusted source sends the $|H\rangle$, and $|V\rangle$ states

Input state	C_H	C_V	C_D	C_A
$\|H\rangle$	6652244	13709	3409198	3550172
$\|V\rangle$	13956	7669421	3694149	3328177

C_H, C_V, C_D, and C_A represent the raw counts of the detectors corresponding to the H, V, D, and A ports, respectively.

It should be noted that our exploited SI-QRNG protocol requires the measurement devices are trusted and well-characterization. A key parameter is the detector efficiency characterization. The calibration of detector efficiency is a highly complex issue that requires consideration of multiple factors, such as the stability of the source and the confidence of the power meter. Given the limitations of our current experimental conditions, achieving the optimal uncertainty in calibration efficiency is challenging[73]. Therefore, in our work, we have adopted the same approach as other existing QRNGs[40,46]-directly assuming that the detector efficiency testing is reliable and accurate, and using the average detection efficiency as our trusted efficiency in the experiment. Furthermore, current advanced commercially available detectors can already provide beyond 60% efficiency at 300 MHz[74–76], higher than the performance of the detector we currently use (1.9% at 8 MHz), thus reducing the impact of efficiency calibration uncertainty.

Characterization of the worst incompatibility between *Z* and *X* basis

Since the exploited protocol is source-independent, the worst incompatibility between the Z and X basis in the decoder chip determines the optimized generation rate obtained by our setup. The process of quantifying the worst incompatibility between the Z and X measurement basis in the decoder chip is the same as the schematic setup in Fig. 2, except that we assume the source is trusted and use an additional polarization controller to generate trusted polarization states. The detailed process is reviewed as follows:

First, we use a trusted laser source to generate polarized light pulses and send these pulses to the decoder chip. Then, we scan the voltages of PS3 and PS4 until the detection count at the D port is maximized while the detection count at the A port is minimized. Next, we scan the voltages of PS1 and PS2 until the detection counts at the H port and the V port are approximately the same. At this stage, the decoder chip is reconfigured in quantum measurement in Z and X basis.

Subsequently, we adjust the polarization controller until a maximum extinction ratio C_H/C_V is obtained. We refer to the input state as $|H\rangle$ and record the corresponding detection counts of the four SPDs. The recorded detection counts at the H port, V port, D port, and A port are denoted as C_H, C_V, C_D, and C_A, respectively. Similarly, we can obtain the corresponding detection counts of the four SPDs for the input state $|V\rangle$. All recorded detection events when the trusted source sends different polarization states are listed in Table 2. Then, we can calculate the incompatibility between the Z and X measurement basis using the following formula:

$$\begin{aligned} &-2\log_2 \max_{X,Z} |\langle X|Z\rangle| \\ &= \min\left\{-\log_2 \frac{\max\{C_D^H, C_A^H\}}{C_D^H + C_A^H}, -\log_2 \frac{\max\{C_D^V, C_A^V\}}{C_D^V + C_A^V}\right\}, \end{aligned} \tag{4}$$

the superscript C indicates the detection counts of the corresponding detector when emitting H and V polarized light. By incorporating the experimental data in the Table 2. The worst incompatibility between the Z and X basis is determined to be 0.927.

Data availability

The data that support the results of this work are available from the corresponding author on reasonable request.

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Author contributions
Y. D., X. H., X. X. and K. W. conceived the basic idea of the project. Y. D. and K. W. designed and developed the experiment. X. H. and X. X. developed the integrated device. Y. D., Z. Z. (Zhengeng Zhao), X. S. and K. W. implemented the experiment and evaluated the data. Z. Z. (Zhenrong Zhang), X. X. and K. W. supervised the research and experiment. Y. D. and K. W. contributed to the writing of the paper, based on discussions with all authors.

Competing interests
The authors declare no competing interests.

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